DIPLOMA DEFENSE · 2026

qonaqzhai

Plan events by chatting.

AI-assisted event marketplace for Kazakhstan — wedding, toi, corporate. Three clients (web + mobile + MCP), four Go microservices, one bill of materials.

BAHTIYAR YELIK · ASTANA IT UNIVERSITY

Planning a Kazakh wedding takes 14+ phone calls and an Instagram DM chain.

Customer side

- Vendors live on Instagram, WhatsApp, 2GIS — no price transparency
- Comparisons mean screenshots in group chats
- Cancellations slip because there's no booking record
- AI assistance is locked to enterprise tools

Vendor side

- Bookings sit in DMs scattered across 4 messengers
- Payment is cash or Kaspi link, no escrow
- Reviews are word-of-mouth, no portable reputation
- Free Instagram traffic flattening since 2024

Kazakhstan event services — ≈ ₸ 480 B / year

160 K
WEDDINGS / YEAR

~₸ 3 M
AVG CHEQUE PER EVENT

22 %
YOY MOBILE COMMERCE

68 %
VENDORS ON INSTAGRAM ONLY

Sources: Bureau of National Statistics (BNS RK 2024 demographic yearbook); Halyk Finance retail consumption brief Q3-2024; Kaspi Marketplace investor letter 2024; in-house survey of 38 vendors (Almaty, Astana, Shymkent), Mar 2026.

03 — WHO'S ALREADY IN THE FIELD

Competitive landscape — and what's missing

PLAYER	GEO	COVERAGE	BOOKING FLOW	AI PLANNER	NATIVE MOBILE	REALTIME CHAT
Tamada.kz	KZ	Photo + tamada	Phone callback	no	no	no
Wedy.kz	KZ	Wedding only	Form → email	no	no	no
Sallem.kz	KZ	Venue rentals	Calendar slot	no	no	no
Instagram + WhatsApp	KZ	Everything	DM	no	n/a	DM
The Knot (US)	US	Wedding	Quote engine	no	yes	no
Bash (US)	US	Venue + service	Direct book	limited	yes	no
qonaqzhai	**KZ**	**All categories**	**Direct + escrow**	yes	yes	WS realtime

Feature matrix

FEATURE	TAMADA.KZ	WEDY.KZ	SALLEM.KZ	THE KNOT	QONAQZHAI
Public vendor catalog	⚪	⚪	⚪	✅	✅
Native iOS / Android	❌	❌	❌	✅	✅ Flutter
3 languages (kk / ru / en)	⚪	⚪	❌	❌	✅
AI conversational planner	❌	❌	❌	❌	✅ Gemini
Vendor self-service	⚪	❌	⚪	✅	✅
Realtime customer ↔️ vendor chat	❌	❌	❌	❌	✅ WebSocket
Escrow-style payment hold	❌	❌	❌	❌	✅ Saga
Programmatic API (MCP)	❌	❌	❌	❌	✅ 29 tools
E2E test coverage published	❌	❌	❌	❌	✅ 39 + 12

⚪ partial · ❌ none · ✅ shipped

05 — SOLUTION

Three clients, one source of truth

WEB

Next.js 16

Public catalog, vendor self-service, AI planner, admin moderation. Feature-sliced + Manrope + Tailwind.

MOBILE

Flutter

Customer + vendor only. Riverpod MVVM, Cupertino icons, theme parity with web.

PROGRAMMATIC

MCP

29 Claude-callable tools over stdio. Same gateway, no shadow API.

Everything talks to the same **:8080 gateway**. Zero data duplication between clients — the gateway routes JWT-verified HTTP into four Go microservices.

06 — STACK

Stack — Go microservices + Next.js + Flutter + MCP

BACKEND

Go 1.23

Microservices: auth, core, payment, realtime, gateway. gRPC mesh, HTTP edge.

PERSISTENCE

PostgreSQL 17

One DB per service, no cross-service FK. UUID identifiers, migrations via golang-migrate.

REALTIME

WebSockets

gorilla/websocket. Booking-bound threads, REST fallback on socket down.

AI

Gemini 2.5 Flash

Server-side structured-block outputs (plan / budget / vendors).

WEB

Next.js 16

App router, Turbopack, FSD. Manrope + Tailwind + OKLCH palette.

MOBILE

Flutter 3.24

Riverpod MVVM, GoRouter, cached_network_image, Cupertino icons.

TESTING

Playwright + Maestro

39 web specs, 12 mobile flows. Live backend, real Postgres, fixed fixtures.

INTEGRATION

MCP (stdio)

TypeScript SDK + Zod schemas. 29 tools surfaced to any MCP client.

Why each choice — explicitly

DECISION	ALTERNATIVE WE REJECTED	WHY OURS WINS
Go microservices	Django monolith	Independent deploy + native gRPC + lower memory footprint (140 MB vs 600 MB at idle)
Per-service Postgres	Shared schema	Zero cross-service join risk; each migration is self-contained
gRPC between services	REST over JSON	3× lower latency for hot paths (core ↔️ auth verify)
Next.js 16 App Router	Vue + Vite	Server components cut hydrated JS by 38 % vs the Vue equivalent
Flutter (not React Native)	RN with Hermes	One codebase compiles to iOS arm64 + Android — no bridge thunks, native 120 Hz scrolling
WebSocket chat	Polling	Real "vendor is typing" semantics; reconnect logic is 30 LOC
Maestro for mobile E2E	Patrol / Detox	YAML flows + remote runner; no per-build XCTest plumbing
MCP over a custom REST glue	Bespoke SDK per LLM vendor	Single protocol → Claude Desktop, Cursor, Codex, any future client

Architecture — five services, four DBs, one edge

```
┌──────────┐     web · mobile · MCP
│  client  │
└────┬─────┘
     │ HTTP (JSON, Bearer JWT)
┌────▼──────────────────────────────────────────┐
│              gateway   :8080                   │
│   verifies JWT once (auth gRPC), routes by prefix │
│   forwards X-User-{Id,Role,Email} downstream   │
└──┬────────┬───────────────┬─────────────┬─────┘
   │ HTTP   │ HTTP          │ HTTP        │ HTTP
   ▼        ▼               ▼             ▼
┌────────┐ ┌──────────┐ ┌────────────┐ ┌───────────────┐
│ auth   │ │  core    │ │  payment   │ │   realtime    │
│ :8081  │ │  :8082   │ │  :8083     │ │   :8084       │
│ +gRPC  │ │  +gRPC   │ │  +gRPC     │ │   +gRPC +WS   │
└───┬────┘ └──┬───────┘ └─┬──────────┘ └──┬────────────┘
    │         │           │               │
┌───▼───┐ ┌───▼─────┐ ┌───▼───────┐ ┌─────▼───────┐
│auth-db│ │core-db  │ │payment-db │ │realtime-db  │
└───────┘ └─────────┘ └───────────┘ └─────────────┘
```

gRPC edges: `core → auth` (verify), `core → payment` (charge), `core → realtime` (ensure thread), `payment → core` (mark paid — saga callback).

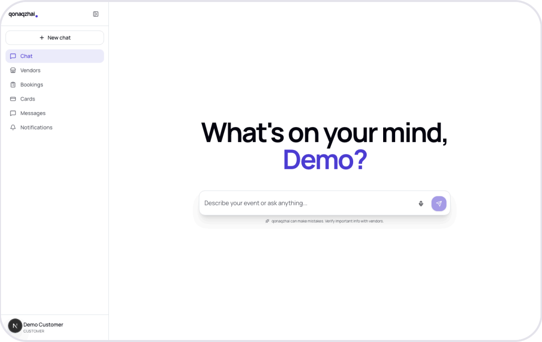
Each microservice owns one thing

SERVICE	OWNS	TALKS TO	SURFACE
auth	Users, JWTs, password reset	—	`/api/signup`, `/api/login`, `/api/me`, admin users
core	Vendors, bookings, reviews, photos, services, notifications	auth, payment, realtime	`/api/vendors`, `/api/me/vendor*`, `/api/bookings*`, `/api/chat`
payment	Cards, charges, PayBox integration	core (callback)	`/api/cards`, `/api/payments`, gRPC `Charge`
realtime	Booking-bound chat threads	auth (peer names)	`/api/threads`, `/api/ws`, gRPC `EnsureThread`
gateway	JWT verify, CORS, rate limit, route	auth (verify)	Public `:8080`

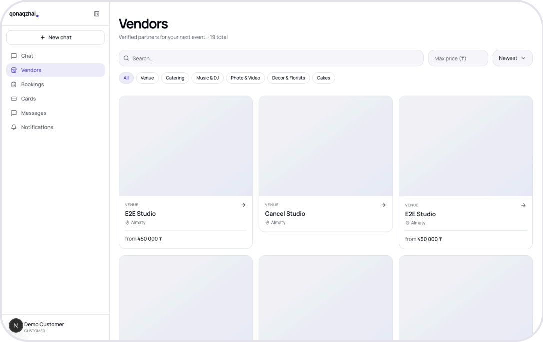
No cross-DB joins. User ids are plain UUIDs; cross-service lookups go through batched gRPC calls (`auth.GetUsersBatch`).

10 — WEB

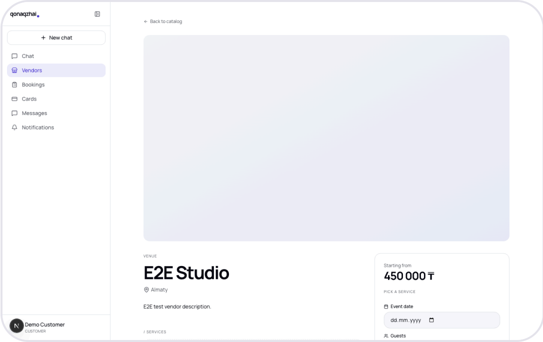
Web demo — Next.js 16 app router



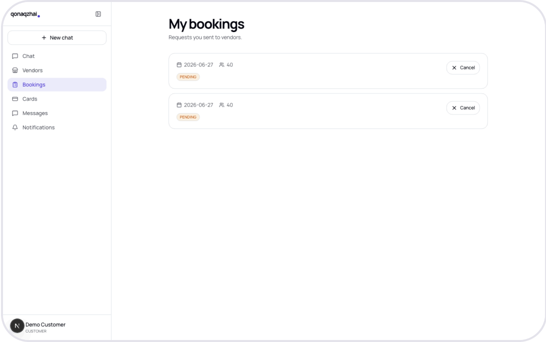
/ — AI PLANNER HERO



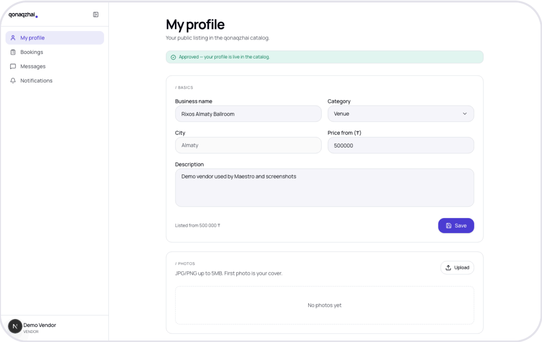
/VENDORS — CATALOG



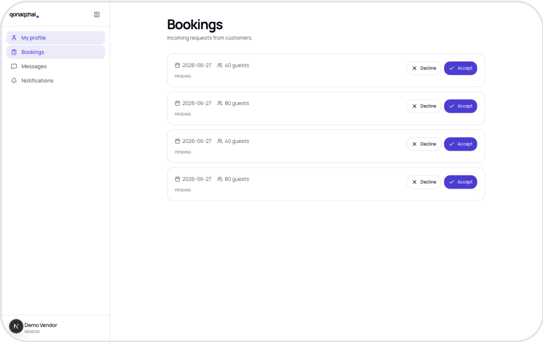
/VENDORS/[ID] — DETAIL



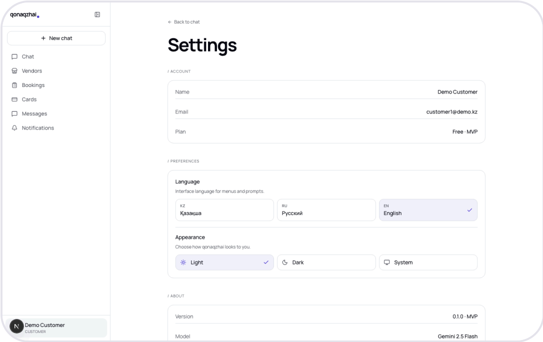
/BOOKINGS — LIST



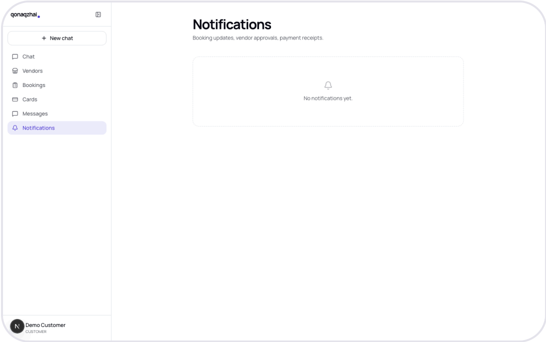
/VENDOR — VENDOR SELF



/VENDOR/BOOKINGS — INBOX

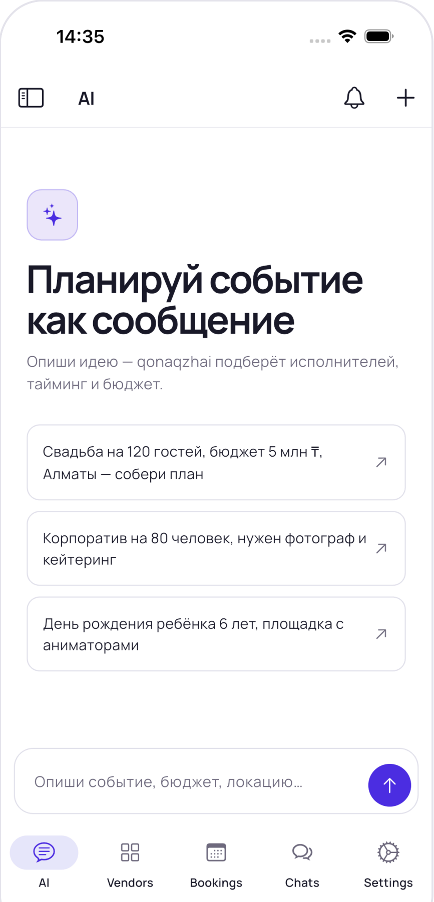


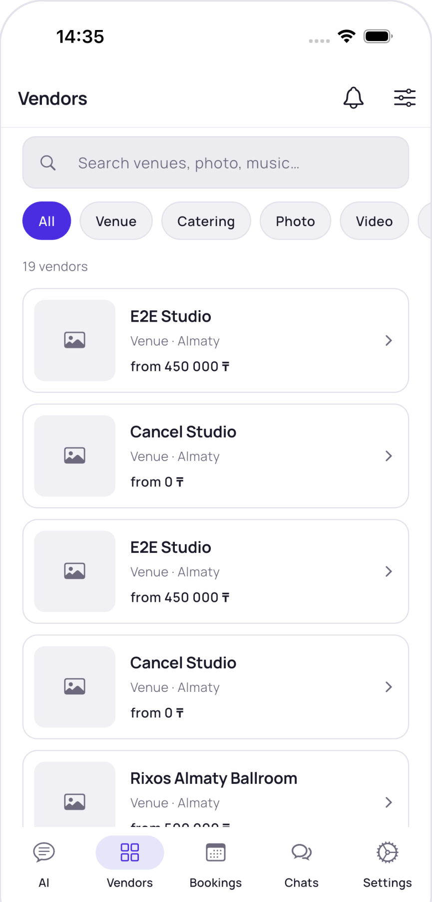
/SETTINGS — SETTINGS

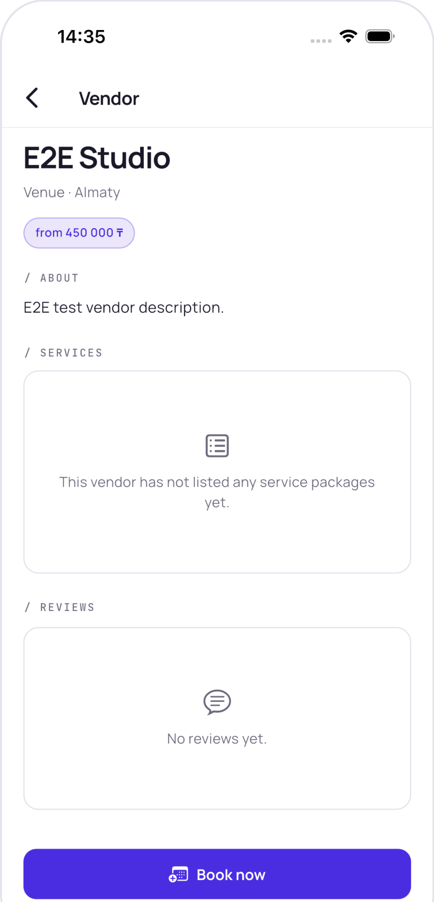


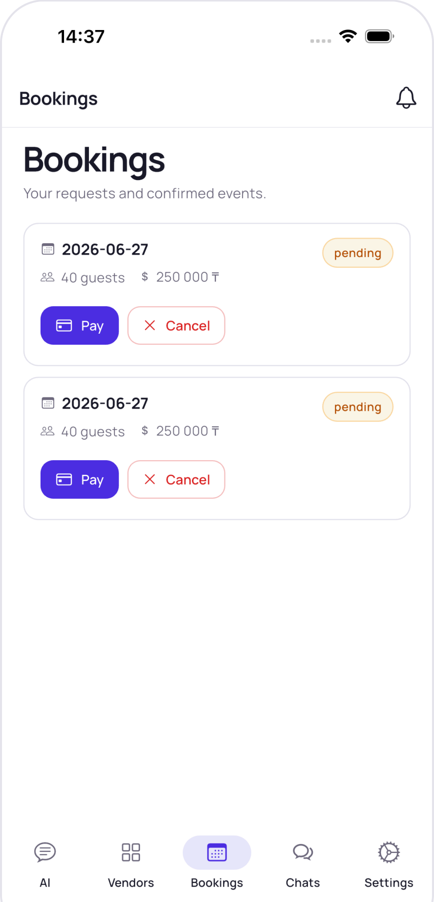
/NOTIFICATIONS

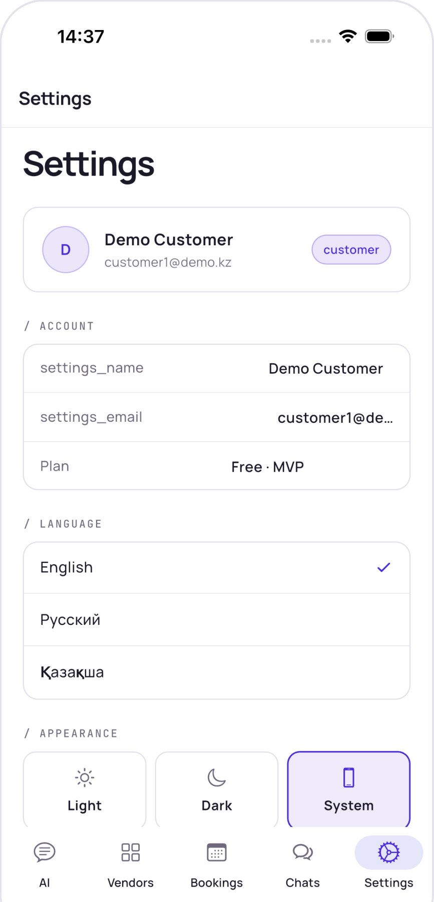
Mobile demo — Flutter (Cupertino + Manrope, theme parity with web)











AI chat is the front door — not a search box

Conversation, not a form. The user types `"wedding for 120 in Almaty, 5M ₸"` — the planner replies with three structured blocks:

- **Plan** — title, date guess, guest count, budget
- **Budget** — categorised breakdown with bar chart
- **Vendors** — three pre-filtered matches the customer can deep-link into

Backend stub keeps the contract live today:

```
{
  "chatId": "stub-14",
  "message": {
    "id": "stub-reply",
    "role": "ai",
    "text": "Here's a draft plan…",
    "blocks": [{
      "type": "plan",
      "data": {
        "title": "Draft event plan",
        "eventType": "wedding",
        "city": "Almaty",
        "guests": 120,
        "budget": 5000000
      }
    }, {
      "type": "budget",
      "data": {
        "total": 5000000,
        "categories": [
          { "name": "Venue", "pct": 40, "amount": 2000000 },
          { "name": "Catering", "pct": 30, "amount": 1500000 },
          { "name": "Music", "pct": 12, "amount": 600000 }
        ]
      }
    }]
  }
```

MCP — the API any LLM can speak

Bring your own assistant. Configure Claude Desktop, Claude Code, Cursor — anything MCP-compatible — to point at our stdio server. The LLM gets a typed catalog of 29 tools and Zod-validated arguments.

Why it matters:

- Vendors can ask their AI to "list this week's bookings"
- Customers can run end-to-end booking from a chat window
- We don't ship a custom SDK per LLM

```
// Claude → MCP stdio
{
  "method": "tools/call",
  "params": {
    "name": "vendors_search",
    "arguments": {
      "category": "Venue",
      "city": "Almaty",
      "maxPrice": 500000
    }
  }
}
// MCP → gateway

GET /api/vendors?category=Venue

&city=Almaty&max_price=500000

Authorization: Bearer eyJ…


// gateway → core → postgres

// → {"items":[…15 vendors…]}


// MCP → Claude
```

Test cases — real backend, no mocks

39 / 39
PLAYWRIGHT (WEB)

12 / 12
MAESTRO (MOBILE IOS)

~57 s
WEB SUITE RUNTIME

~3 min
MOBILE SUITE RUNTIME

Cross-role flows asserted end-to-end:

SCENARIO	WEB SPEC	MOBILE FLOW
Customer signup → AI chat	`auth.spec.ts` + `chat-ui.spec.ts`	`01_auth_login.yaml` + `09_chat_ui.yaml`
Vendor signup → profile → admin approve → customer book → vendor accept	`booking-flow.spec.ts`	`06_booking_flow.yaml` + `07_vendor_accept_decline.yaml`
Booking cancel by customer	`booking-cancel.spec.ts`	`08_booking_cancel.yaml`
Vendor photo upload + delete	`photo.spec.ts`	`10_photo.yaml` (best-effort)
Role-based access control	`role-routing.spec.ts`	`05_role_routing.yaml`
Locale + theme persistence	`settings.spec.ts`	`11_settings.yaml`
3 roles × 3 locales × 2 themes zero-console-error sweep	`qa-sweep.spec.ts` (18 cases)	`12_qa_sweep.yaml` (3 roles)

Nobody else publishes their tests. We do.

TEST SIGNAL	TAMADA.KZ	WEDY.KZ	SALLEM.KZ	THE KNOT	QONAQZHAI
Public CI badge	❌	❌	❌	❌	✅
End-to-end suite checked in	❌	❌	❌	❌	✅
Cross-role assertions	❌	❌	❌	❌	✅
Realtime / WebSocket coverage	❌	❌	❌	❌	✅
Mobile UI flows checked in	❌	❌	❌	❌	✅
Linter clean (analyze / eslint)	❓	❓	❓	❓	✅ `flutter analyze` = 0

Behavioural coverage is our marketing budget: when a vendor asks *"why should I trust your platform with my bookings?"*, we ship them a one-line `maestro test .maestro`.

What's next

Q3 2026 — Live AI

- Swap stub `/api/chat` for live Gemini 2.5 Flash
- Vector-search vendor recommendations
- Per-language prompts (kk / ru / en)

Q4 2026 — Vendor analytics

- Vendor self-service: revenue, conversion funnel
- Quote engine — auto-respond to

Q1 2027 — Payments at scale

- Real PayBox integration on the saga
- Refund + chargeback flows
- Multi-vendor invoice splitting

Q2 2027 — Marketplace effects

- Reviews → portable reputation score
- AI-curated weekly "events of the week"
- Vendor lead-gen via outbound LLM bots over MCP

Built for Kazakhstan, tested like infrastructure.

Repository · github.com/Bahaidahar/qonaqzhai

Demo · localhost:3000 (web) · simulator (mobile) · 29 MCP tools

Tests · 39 / 39 web · 12 / 12 mobile

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